EXPERIMENT – 5

Performing Various Configurations of IPTables Firewall in Kali Linux

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Open Terminal and Check IPTables

1. Open **Terminal** in Kali Linux.
2. Check IPTables version:
3. iptables --version

Step 3: View Existing Firewall Rules

1. Display current IPTables rules:
2. sudo iptables -L

Step 4: Set Default Policies

1. Set default policy to drop incoming traffic:
2. sudo iptables -P INPUT DROP
3. Allow outgoing traffic:
4. sudo iptables -P OUTPUT ACCEPT

Step 5: Allow Localhost Traffic

1. Allow loopback interface traffic:
2. sudo iptables -A INPUT -i lo -j ACCEPT

Step 6: Allow SSH Access

1. Allow SSH connections on port 22:
2. sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT

Step 7: Allow HTTP and HTTPS Traffic

1. Allow HTTP traffic:
2. sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
3. Allow HTTPS traffic:
4. sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT

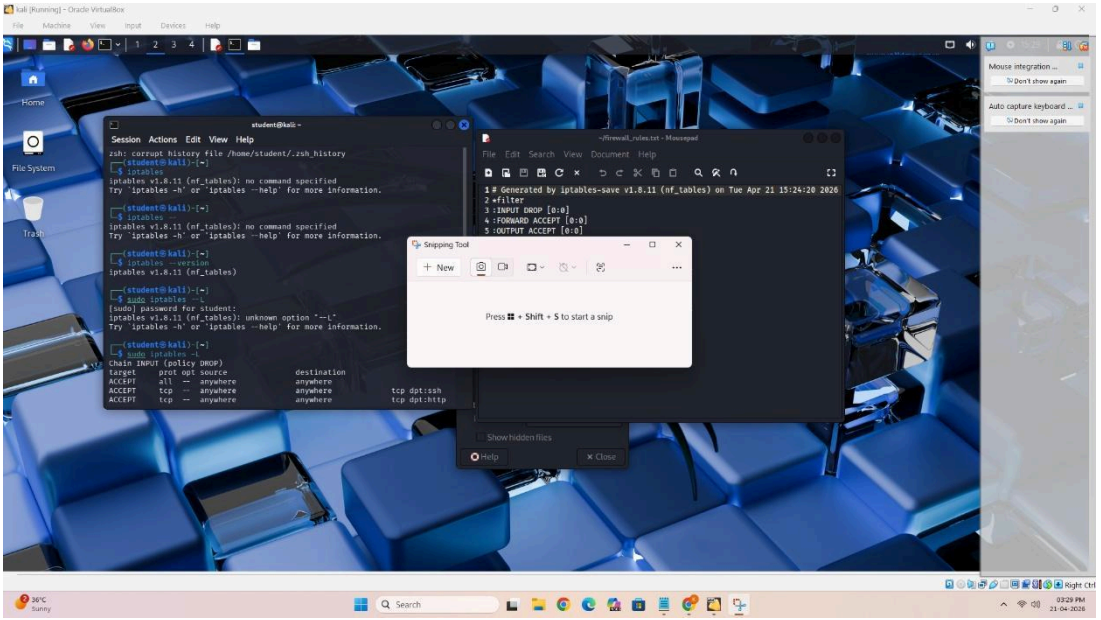
Step 8: Block a Specific IP Address

1. Block traffic from a specific IP:
2. sudo iptables -A INPUT -s 192.168.1.100 -j DROP

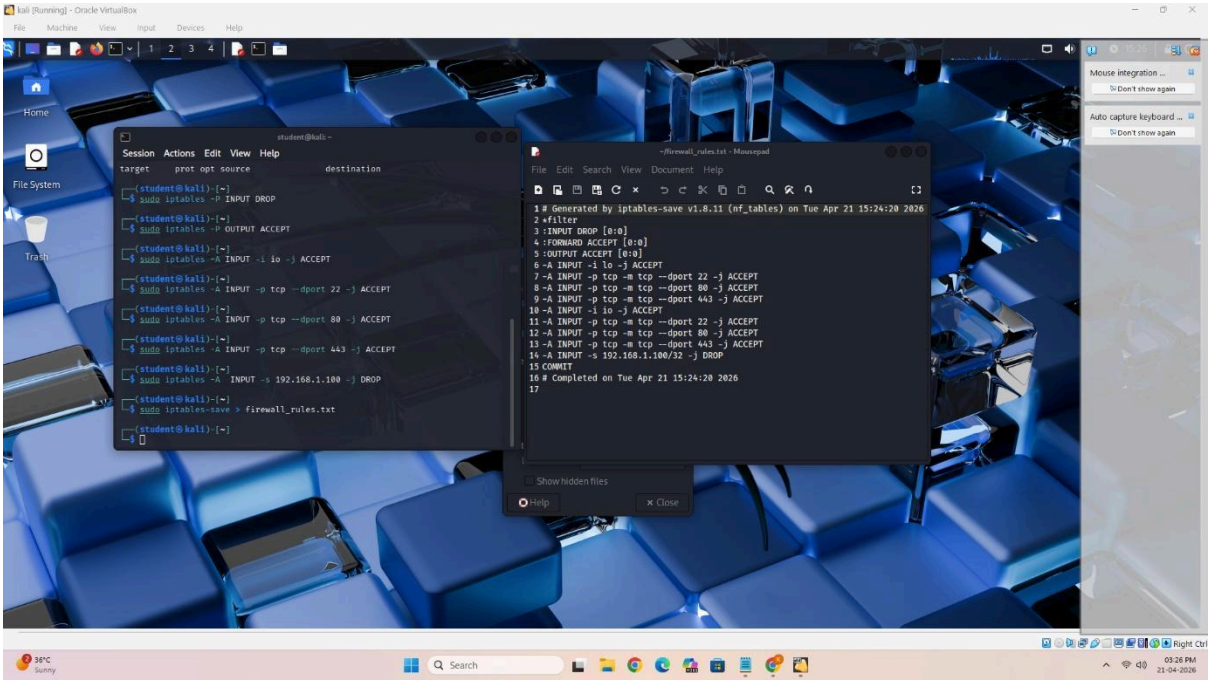
Step 9: Save IPTables Rules

1. sudo iptables-save > firewall_rules.txt

Output




```
└─$ iptables --version
iptables v1.8.11 (nf_tables)

┌──(student㉿kali)-[~]
└─$ sudo iptables --L
[sudo] password for student:
iptables v1.8.11 (nf_tables): unknown option "--L"
Try `iptables -h' or 'iptables --help' for more information.

┌──(student㉿kali)-[~]
└─$ sudo iptables -L
Chain INPUT (policy DROP)
target     prot opt source               destination
ACCEPT     all  --  anywhere             anywhere
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:ssh
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:http
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:https

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination

┌──(student㉿kali)-[~]
└─$ sudo iptables -P INPUT DROP

┌──(student㉿kali)-[~]
└─$ sudo iptables -P OUTPUT ACCEPT

┌──(student㉿kali)-[~]
└─$ sudo iptables -A INPUT -i io -j ACCEPT

┌──(student㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT

┌──(student㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT

┌──(student㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT

┌──(student㉿kali)-[~]
└─$ sudo iptables -A  INPUT -s 192.168.1.100 -j DROP

┌──(student㉿kali)-[~]
└─$ sudo iptables-save > firewall_rules.txt
```



Result

Various firewall configurations were successfully implemented using IPTables in Kali Linux. The experiment demonstrates how firewall rules can be used to control network traffic and improve system security.